SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: **Database Management Systems**

COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : MCQ Test (Low)

Assessment Tool Weightage: 30%

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	**5**
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	**5**
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	**5**
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	**5**
5	Functional dependency X → Y means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	**5**
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	**5**

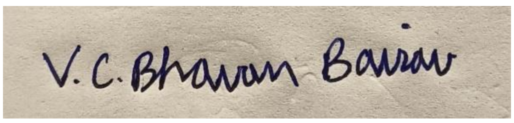
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key reference	BL2 – Understand	**5**
8	If R(A,B,C) has FD: A→B and B→C, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	**5**
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	**5**
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	**5**

Code of Conduct:

I <u>V.C.BHAVAN BAIRAV – 192511369</u> (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

MCQ					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1. Which of the following is true about 1NF?

   ANSWER: (c) All attributes must be atomic

   JUSTIFICATION: 1NF requires every attribute to contain a single, indivisible value and does not allow repeating groups or multi-valued attributes.

2. A relation is in 2NF if it is in 1NF and:

   ANSWER: (b) Has no partial dependencies on primary key

   JUSTIFICATION: 2NF eliminates partial dependencies, meaning non-key attributes must depend on the entire primary key.

3. Which normal form deals with multi-valued dependencies?

   ANSWER: (d) 4NF

   JUSTIFICATION: 4NF specifically addresses non-trivial multivalued dependencies and removes redundancy caused by independent multi-valued attributes.

4. BCNF is a stricter version of:

   ANSWER: (c) 3NF

   JUSTIFICATION: BCNF strengthens 3NF by requiring every determinant in a non-trivial functional dependency to be a candidate key.

5. Functional dependency $X \rightarrow Y$ means:

   ANSWER: (b) X determines Y

   JUSTIFICATION: If two tuples have the same value of X, they must have the same value of Y. Thus, X functionally determines Y.

6. Which of the following anomaly is NOT associated with unnormalized relations?

   ANSWER: (d) Retrieval anomaly

   JUSTIFICATION: The standard anomalies caused by redundancy are insertion, update, and deletion anomalies; retrieval anomaly is not one of them.

7. A superkey is:

   ANSWER: (b) Any set of attributes that uniquely identifies tuples

   JUSTIFICATION: A superkey can contain one or more attributes that uniquely identify each tuple. It does not have to be minimal.

8. If R(A,B,C) has FD: A→B and B→C, then C is:

   ANSWER: (c) Transitively dependent on A

   JUSTIFICATION: Since A → B and B → C, A indirectly determines C through B. Therefore, C is transitively dependent on A.

9. 5NF is also known as:

   ANSWER: (c) Project-Join Normal Form

   JUSTIFICATION: 5NF is called Project-Join Normal Form (PJ/NF) because it deals with join dependencies and ensures decompositions cannot be further non-trivially decomposed.

10. Lossless-join decomposition ensures:

    ANSWER: (a) No data is lost during decomposition

    JUSTIFICATION: A lossless decomposition allows the original relation to be reconstructed exactly by joining the decomposed relations, without losing or adding incorrect information.